

# K.T.G.Tharusha Nimesh Dhananjaya

B.Sc. Electronics & Telecommunication Engineering (Undergraduate)

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## Education

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University of Moratuwa, Sri Lanka

2024 – present

B.Sc. Eng. Hons. in Electronic & Telecommunication Engineering

- **CGPA** – 3.99/4.00
- **Dean's List** – 1, 2, 3, 4 Semesters
- **Coursework** – Electronics, Signals and Systems, Control Systems, Digital Design, Telecommunication, Data Structures and Algorithms, Computer Vision, Pattern Recognition, Linear Algebra

Maliyadeva College, Kurunegala

2020 – 2022

G.C.E. Advanced Level Examination 2022(2023) – Physical Science Stream

- Combined Mathematics, Physics, Chemistry – 3A's
- Z Score – (+2.382) / District Rank – 27 / Island Rank – 352

Wayamba Royal College, Kurunegala

2014 – 2019

G.C.E. Ordinary Level Examination 2019

- Buddhism, Sinhala Language & Literature, Mathematics, Science, English, History, Tamil, Sinhala Literature, Information and Communication Technology – 9A's

## Courses & Certifications

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1. Sri Lanka Robotics Challenge (SLRC) 2026 – Finalist
2. Machine Learning Specialization – Coursera (Offered by DeepLearning.AI + Stanford University)
  - **Coursework** – Supervised Machine Learning (Regression and Classification), Advanced Learning Algorithms (Neural Networks), Unsupervised Machine Learning, Recommender Systems and Reinforcement Learning.
3. Deep Learning Specialization – Coursera (Offered by DeepLearning.AI) In Progress
  - **Coursework** – Neural Networks, Hyperparameter Tuning, Regularization and Optimization, Convolutional Neural Networks, Sequence Models

## Skills

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- **Languages** – Sinhalese, English
- **Programming Languages** – Python, C, C++, MATLAB, JavaScript
- **Hardware** – System Verilog HDL, FPGA, STM32, Arduino, PIC
- **Machine Learning Libraries** – NumPy, Scikit-learn, PyTorch, TensorFlow & Pandas
- **CAD & EDA Tools** – Solidworks & Altium PCB Designer
- **Documentation** – LaTeX

## Projects

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### **Wearable ECG Monitoring System**

July 2026

(Electronics, IoT, Signal Processing, CNN, Embedded Systems)

- Developed a compact wearable ECG device capable of long-term ambulatory cardiac monitoring, with a focus on early detection of abnormalities in high-risk individuals.
- Built an embedded analytics pipeline that combines real-time ECG signal conditioning with CNN-based pattern recognition to support faster clinical assessment.
- Integrated high-efficiency synchronous buck-boost topologies and precision LDO regulation to maximize signal quality and maintain high analog input impedance.

### **Atom – Robotics Design**

December 2025

(Robotics, Control Systems, Embedded Systems, C++)

- Designed and developed an autonomous mobile robot capable of line and dashed-line following, grid navigation, arrow following, barcode reading, color recognition, wall following, ramp traversal, and autonomous object manipulation for a multi-stage robotics competition.
- Developed the robot control software in C++ on an Arduino Mega platform, implementing PID-based line-following and wall-following controllers and integrating multiple sensors, stepper motors, servo mechanisms and custom PCB electronics.

### **UART Transceiver Implementation in FPGA**

April 2026

(FPGA, Verilog)

- Designed and implemented a UART transceiver in Verilog HDL on FPGA, including RTL development of transmitter, receiver, and top-level modules for asynchronous serial communication.
- Developed comprehensive testbench for functional verification and validated the design on FPGA hardware through successful bidirectional communication between two FPGA boards.

### **Three Channel Audio Mixer with Equalizers**

December 2025

(Analog Electronics, Audio Signal Processing, PCB Design)

- Designed and implemented a compact three-channel analog audio mixer with independent 3-band equalizers (bass, mid, treble), utilizing low-noise op-amps, active Sallen–Key filters, and a Class-AB audio power amplifier.
- Developed multi-board PCBs and enclosure designs with noise-reduction techniques, power-supply isolation, thermal management, and hardware testing to ensure reliable low-distortion audio performance.

### **Automated Switch for Clothing Iron**

June 2025

(Embedded Systems, PCB and Enclosure Design)

- Designed and developed a smart auto-off socket for clothing irons that detects user inactivity and automatically disconnects AC power.
- Designed a PIC microcontroller-based embedded controller integrated with an opto-isolated TRIAC power stage for automated and safe switching of high-voltage AC loads.

### **PWM-Controlled DC-DC Buck Converter**

May 2026

(Power Electronics, Control Systems, Analog Electronics)

- Designed and implemented a PWM-controlled DC–DC buck converter (15 V input, 3–12 V adjustable output, 1 A load current) including MOSFET switching stage, LC output filter, PWM generation circuitry and closed-loop feedback compensator for output voltage regulation.
- Analyzed converter performance using load regulation, line regulation, and efficiency testing.

- Designed and simulated a scalable university campus network and departmental LAN using Cisco Packet Tracer, implementing a three-tier architecture with VLAN segmentation, IPv4 addressing, DHCP, OSPF routing, and Spanning Tree Protocol (STP).
- Developed a fault-tolerant hybrid mesh-ring network topology for the University of Moratuwa, incorporating redundancy, inter-VLAN routing, and wireless access infrastructure with successful connectivity and routing verification through simulation.

## References

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### 1. Dr. Upeka Premaratne

B.Sc. Eng.(Moratuwa), M.E.Sc. (Western Ontario), Ph.D (Melbourne), LL.B. (OUSL), Attorney-at-Law

Senior Lecturer - Grade I,

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### 2. Dr. Samiru Gayan

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